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New paleomagnetic Mesozoic data from the Algarve (Portugal): fast rotation of Iberia between the Hauterivian and the Aptian

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Abstract

We sampled nine sites of Jurassic and Cretaceous limestones in the Algarve region, Portugal. Our study yields a paleopole for the Portlandian–Valanginian ($\phi = 256°$, $\lambda = 58°$, $A_{95} = 5°$) and a paleopole for the Aptian–Albian ($\phi = 232°$, $\lambda = 75°$, $A_{95} = 8°$). These results allow better constraint of the relative rotation of Iberia with respect to Europe: this rotation of $22° \pm 14°$ is a rather fast event occurring during the Barremian, between 132 and 124 Ma. Confinement of Iberia between Africa and Eurasia would explain the rapid rotation of this microplate. The Iberian plate was probably attached to Europe until 132 Ma, then detached from Europe and acted as an independent plate during the time of fast rotation preceding the oceanic spreading in the Bay of Biscay.

Keywords: Iberian peninsula; Mesozoic; paleomagnetism; plate rotation

1. Introduction

Magnetic anomalies of the Atlantic Ocean constrain the relative motions of Eurasia and Africa with respect to North America. These anomalies also yield insight into the kinematic relationships between the Eurasian, African and Iberian plates and constrain the evolution of the Tethys ocean. For example, rifting between Eurasia, Africa, and North America began during the Late Jurassic [1,2]. The rifting phase was accompanied by the formation of several microplates, including Iberia. Some time during the Cretaceous, plate trajectories reorganized and Africa began to converge with Eurasia. The Tethys had virtually disappeared in the Tertiary as a result of the convergent movement of Africa and Eurasia. Correlations of the position of the continent–ocean margins and the identification of anomaly 34 situated in the middle of the Gulf of Biscay help constrain the paleogeographic reconstructions of Iberia [3–5]. However, because a great part of the relative motion of Iberia occurred during the Cretaceous quiet magnetic period, its kinematics are poorly constrained, which leads to large uncertainties in the initial reconstruction.

Such complex tectonic activity can also be moni-

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tored by paleomagnetic studies. This is the case of Iberia, for which paleomagnetic studies suggest it rotated approximately 35° counterclockwise with respect to Eurasia between the Permo–Triassic and the Paleogene [6–10]. Van der Voo [11] presents a synthesis of existing paleomagnetic data and lists expected paleolatitudes and paleodeclinations as a function of time for Madrid if Iberia had remained rigidly attached to either Stable Europe or West Gondwana. He observes that paleolatitude curves for Iberia, Europe and Africa are in very good agreement. Although the paleodeclination curve of Iberia follows that of Africa, it is less well defined. The total amount of rotation of Africa is 40° or more when compared to Europe, while that of Iberia is 30–35°.

Pre-existing paleomagnetic studies in Iberia include Oxfordian results from the Iberian range [12,13], Hauterivian–Barremian and Aptian results from the region around Lisbon [14] and Barremian–Aptian results from Maestrazgo [15]. These studies suggest that: (1) The main phase of rotation of Iberia with respect to Europe occurred between 135 and 120 Ma, while the rotation of Africa with respect to Europe continued until the Tertiary. (2) Before the Middle–Late Jurassic, there is no paleomagnetic evidence for significant motion of Iberia with respect to Eurasia. (3) After 120 Ma, the main phase of rotation was followed by a ''translation'' motion (rotation around a distant Euler pole) during the initial phase of the opening of the Bay of Biscay. The translation ended during the onset of oceanic anomaly 34 in the Late Cretaceous.

Upper Jurassic and lowermost Cretaceous rocks from the region around Lisbon have not yielded reliable results because of remagnetization, which is probably linked to Early Cretaceous rifting. Similarly, the Iberian Range was partially remagnetized during the Early Cretaceous [13,15]. In this paper, we report Jurassic and Cretaceous paleomagnetic data from stratigraphic horizons deposited during the opening of the Atlantic and the Gulf of Biscay. We selected favourable Late Jurassic–Cretaceous sedimentary targets, such as small rift basins that contain well dated limestones series. Because of the effects of Tertiary tectonics, Pyrenean and Betics areas were disqualified. The remaining Late Jurassic–Early Cretaceous targets, which were sampled for this study, include western Portugal, to the north and south of Lisbon, and the Algarve (Fig. 1).

2. Geological framework of the Algarve

The sedimentary basin in the Algarve region of southern Portugal represents an emerged part of the continental margin that formed during the displacement of Africa with respect to Eurasia and Iberia. Mesozoic layers crop out over 150 km in the Algarve basin [16,17] from the cape of Sao Vicente, in the west, to Tavira, in the east, forming a belt whose maximum width is 30 km (Fig. 1). To the north of the central part of the Algarve, the sedimentary facies are continental. To the west, the Jurassic and the lower Cretaceous have continental affinities. The Aptian is represented by a thick (300–500 m) marine section (sections ''Salema'' and ''Almadena–Porto de Mos''). To the east, one finds the thickest lower Cretaceous marine series, particularly in sections called ''Estoi–Pao Branco'', ''Monchara–Pocho'' and ''Alfandanga–Marin''. This section is overlain by Valanginian sandstones, microconglomerates and reefal limestones. After a transgression in the late Valanginian, a regressive period commenced. This period is marked by sedimentary layers that are entirely dolomitised. The lower Aptian is again purely marine and relatively thick. The upper Aptian and the lower Albian are composed of marly calcareous beds, which, in turn, are overlain by another dolomitised series. In the Tertiary, during convergence between Africa and Eurasia, this basin was tilted, uplifted and preserved from intense deformation by the collision, unlike the Betics chains farther east.

In the western Algarve, we sampled one site each of the Albian and Portlandian. In the eastern Algarve, we sampled two Aptian sites, one Albian site, three Valanginian sites and one Portlandian site (Fig. 1). These latter sites are part of the same series from which Galbrun [18] performed a magnetostratigraphic study of the Jurassic/Cretaceous boundary. Galbrun found that the rocks had weak magnetic intensities and contained large amounts of altered magnetic minerals. Despite his findings, because this sequence is very complete and continuous from the end of the Jurassic to the upper Cretaceous, we thought it useful to sample on either side of his

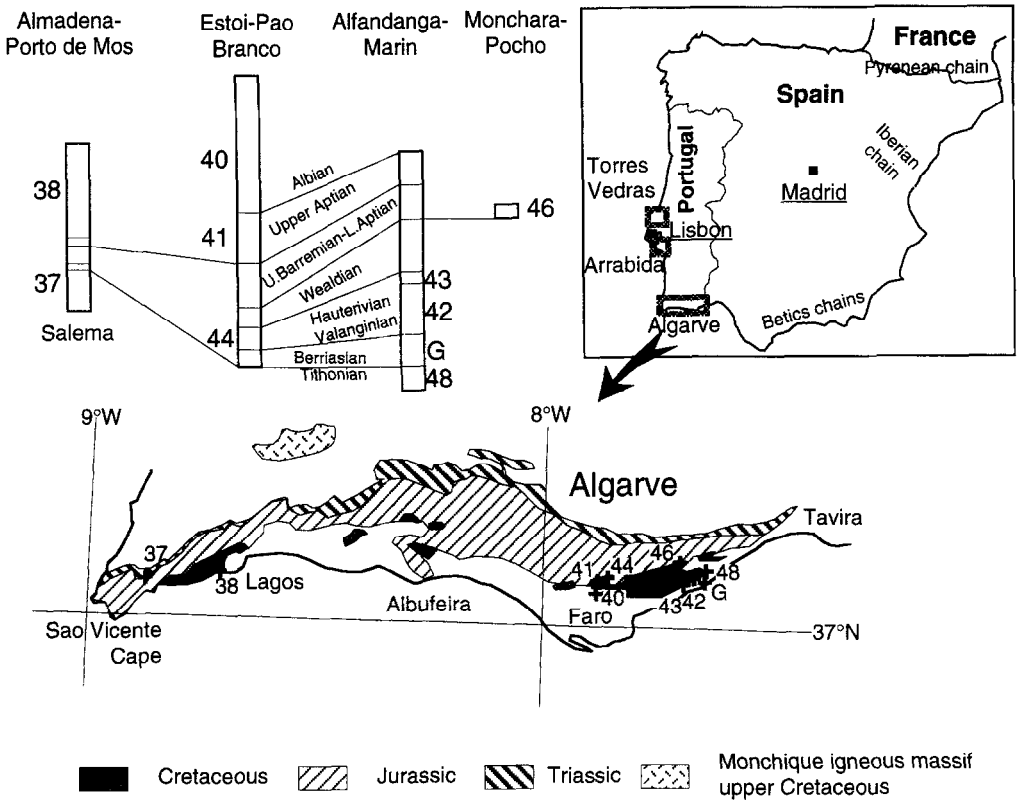


Fig. 1. Simplified geological map of the Algarve with the location, number and statigraphy of each sampling site. *G* indicates a magnetostratigraphy of the Jurassic–Cretaceous boundary [18]. Stratigraphic sections that have been sampled are given in the upper left. Inset: location of the study zones.

sample locality. Horizons which were dolomitized were not sampled because they are often unsuitable for paleomagnetism.

3. Paleomagnetic study

3.1. Magnetic carriers

Following the method described by [19], a composite IRM using magnetic fields of 1.2, 0.4 and 0.12 T were successively applied to each of the three perpendicular axes of a sample prior to thermal demagnetization. This method facilitated a more accurate identification of magnetic carriers (Fig. 2).

For the Portlandian (sites 37 and 48; Fig. 2a,b), Aptian (sites 38 and 46, Fig. 2d,e) and Albian (site 40, Fig. 2f) the experiment demonstrates the presence of two magnetic carriers: one that is not saturated by fields of 1.2 T and whose unblocking temperature occurs at 120°C, and another that has weak coercivities and unblocking temperatures up to 500°C. The former most likely corresponds to goethite, while the latter, which is the main carrier of the remanence, probably corresponds to titano-magnetite. A weak fraction of titano-magnetite is also found in Valanginian sites 42, 43 and 44 (Fig. 2c). For the three Valanginian sites 42, 43 and 44 (Fig. 2c), and one Aptian site 46 (Fig. 2e) magnetic carriers with high coercivity are also found. Their magnetization is destroyed only above 625°C. Taken together, the experiments suggest hematite resides in these samples. Another magnetic carrier is found with medium (site 48, Fig. 2b) or hard (site 38, Fig. 2d) coercivities and unblocking temperatures in the range 250–300°C. These characteristics are likely attributable to pyrrhotite.

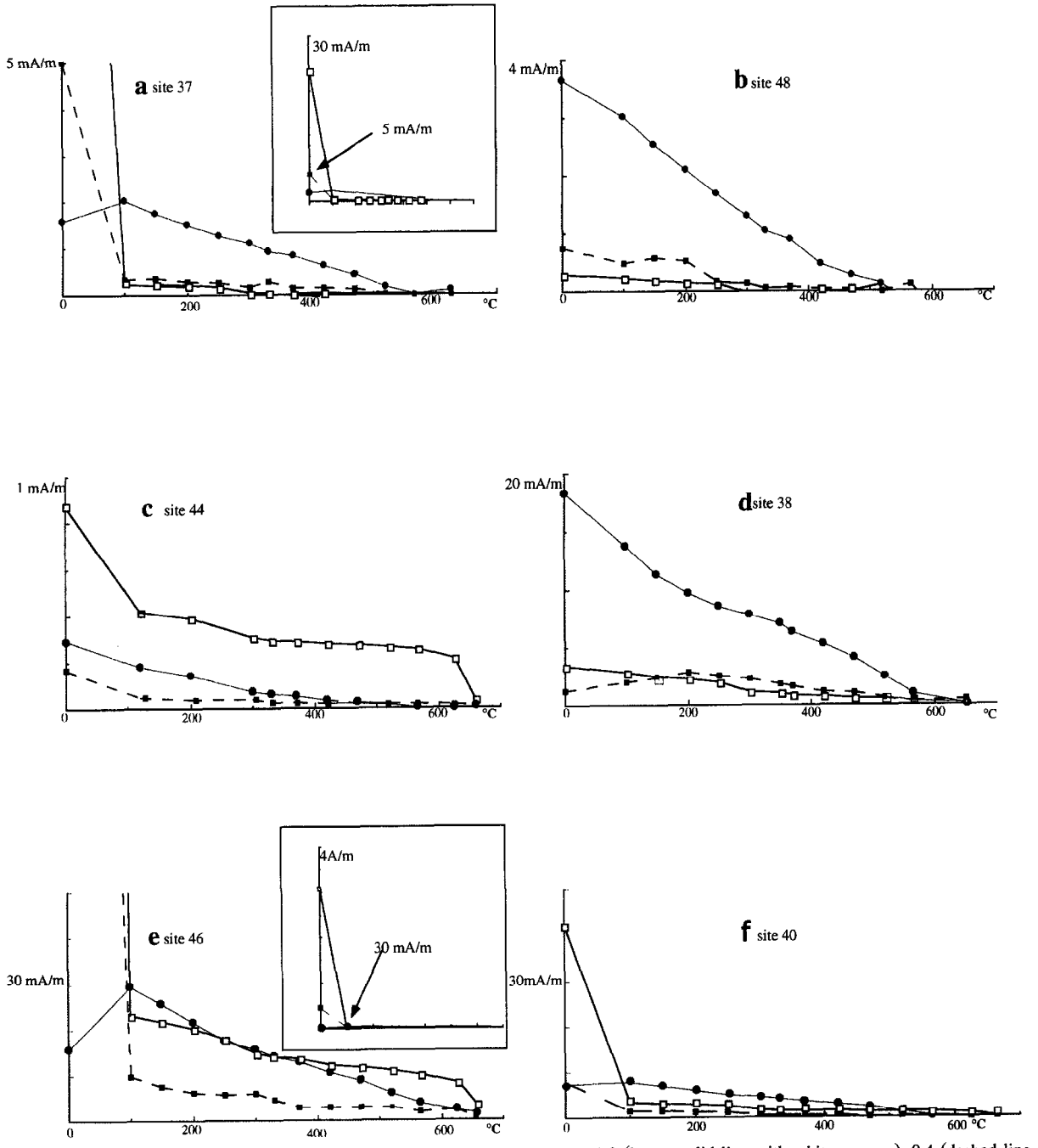


Fig. 2. Thermal demagnetization of a composite IRM using magnetic fields of 1.2 (heavy solid line with white squares), 0.4 (dashed line with black squares) and 0.12 T (lighter solid line with dots) were successively applied to each of the three perpendicular axes [19]. Six characteristic samples are represented: (a) and (b) Jurassic sites; (c) Valanginian site; (d) and (e) Aptian sites; (f) Albian site. For very weathered samples (a) and (e), the demagnetization stage of the goethite at 120°C is drawn in a frame.

3.2. Paleomagnetic measurements

All remanence measurements were made with a 3-axis CTF cryogenic magnetometer housed in a shielded room of the paleomagnetic laboratory at the Institut de Physique du Globe de Paris. Specimens were subjected to stepwise alternating field demagnetization or stepwise thermal demagnetization. Stepwise AF demagnetization was performed using a Schonstedt alternating field demagnetizer and stepwise thermal demagnetization was conducted using a magnetically shielded oven whose residual magnetic field is lower than 10 nT. The characteristic direction was determined by a least-squares straight line, constrained to pass through the origin [20].

The samples exhibit a significant decrease (30–90%) of intensity of natural remanent magnetization (NRM) as soon as they are heated to 120°C. Taking this into account, all samples were heated to 120°C then further demagnetized by either heating or applying an alternating magnetic field. Results are presented in upward stratigraphic order (see also Fig. 1).

3.2.1. Portlandian (Figs. 3 and 4, Table 1)

Site 37: Despite the weak intensity of magnetization (about 10^{-5} A/m) after heating the samples to 120°C, a characteristic direction can be isolated for most of the samples, that is defined between 270° and 450–500°C or by alternating fields between 20 and 80 mT. Above 500°C the magnetization becomes unstable.

Site 48: The weathering is variable and is indicated by the colour of the samples, which ranges from yellow–beige to grey–blue. The intensity of magnetization ranges between 10^{-5} and 10^{-4} A/m. Among the 15 samples studied, most are demagne-

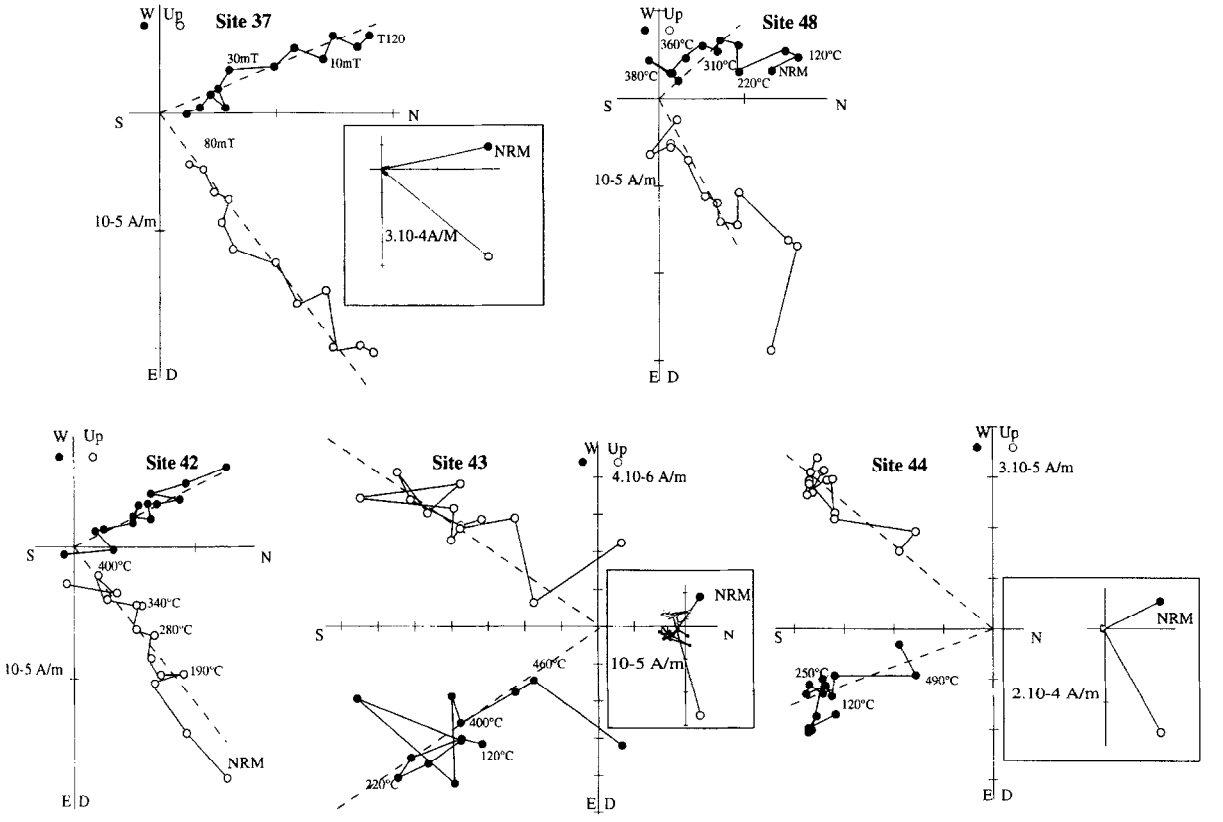


Fig. 3. Typical demagnetization diagrams of one specimen from each Jurassic and lower Cretaceous site. Orthogonal projection of the remanent magnetization vector on the horizontal (dots) and vertical plane (N–S, circles), after tectonic correction. Temperature in °C and intensity of the alternative field in mT. For weathered samples the first stage of demagnetization is shown framed.

tized below 350°C and define a direction which is close to the present dipole earth field (PEF). However, 6 samples contain a direction of magnetization still well defined by 400°C that is distinct from the PEF. These latter directions have been retained.

3.2.2. Valanginian (Figs. 3 and 4, Table 1)

Sites 42, 43 and 44: Of the 45 samples demagnetized from these sites, 21 contain solely the PEF direction. After heating the samples to 120°C, the intensity of magnetization ranges between 5×10^{-5} and 5×10^{-6} A/m.

Site 42: The characteristic direction is unblocked between 180° and 380–450°C. Above these temperatures, the magnetization increases and becomes unstable. We have not identified reversed directions in this site.

Site 43: The characteristic direction is defined between 350°C and 520°C. For some samples, the magnetization is not completely removed before the directions become unstable, thus their characteristic directions were calculated by means of Fisher statistics. For other samples, it is not possible to remove the PEF completely. In this case, remagnetization great circles were used to calculate the characteristic directions [21].

Site 44: These rocks are much more altered than in the two previous Valanginian sites and they become unstable at lower temperatures. Consequently, for several samples among the 15 studied, directions had to be determined by remagnetization circles.

Directions determined for sites 43 and 44 are mainly of reversed polarity and the mean directions are antipodal to that of site 42 at the 95% confidence level.

3.2.3. Aptian (Figs. 5 and 6, Table 1)

Site 38: NRM intensities range between 1 and 3×10^{-4} A/m. We have identified two components, one that is destroyed around 300°C (38lT, Table 1), and the other that has a broad unblocking spectra up to 500°C (38hT, Table 1). The low temperature component (LTC) gives a more westward direction than the high temperature component (HTC). The

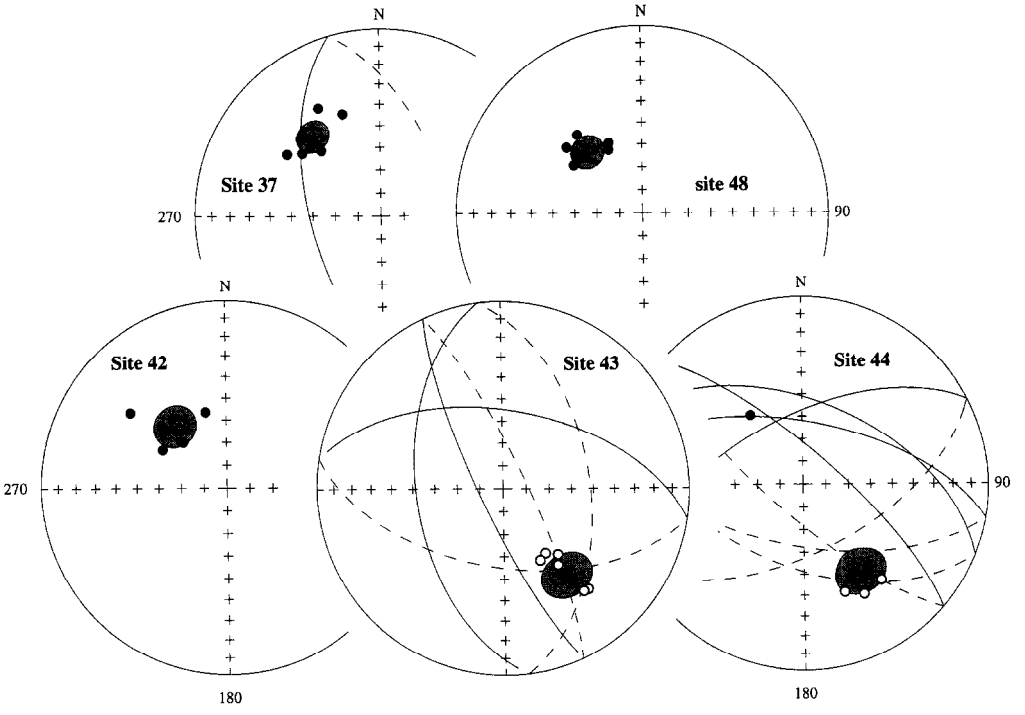


Fig. 4. Equal-area projection of characteristic directions after tectonic correction and the mean direction (star with greyed confidence circle) for each Jurassic and Lower Cretaceous site. Solid symbols = down directions; open symbols = up directions.

Table 1
Paleomagnetic results

No.	Stratigraphic age	Means before tectonic correction			Means after tectonic correction					
		n/N	Dg°	Ig°	Ds°	Is°	k	$\alpha 95°$		
37	Portlandian	9/13	307.7	48.0	317.6	42.7	66	6.4		
48	Portlandian	6/15	336.0	49.1	324.3	50.0	87	7.2		
G	Berriasian	40	341.8	46.4	328.0	45.9	32	3.9		
42	Valanginian	7/15	341.3	43.5	320.9	51.2	42	9.4		
43	Valanginian	9/15	157.2	−38.8	144.9	−45.5	41	8.4		
44	Valanginian	8/15	164.8	−44.5	147.3	−43.2	38	9.4		
38 lT	Aptian	9/11	332.3	42.9	336.0	46.1	28	9.9		
38 hT	Aptian	8/11	346.3	46.1	351.2	48.2	32	10.0		
41	Aptian	7/11	342.2	27.7	335.9	45.9	38	9.9		
46	Aptian	11/15	350.3	68.8	352.6	45.8	67	5.6		
40	Albian	9/11	349.3	25.5	343.0	50.3	32	9.4		
Mean values		N					k	$\alpha 95°$	k_s/k_g	Pole position
										Φ λ
37, 48	Upper Jurassic	2	321.7	49.4			–	–		
					320.7	46.4	–	–		
G, 42, 43, 44	Lowest Cretaceous	4	341.2	43.3			433	4.4		
					325.4	46.5	408	4.6	1	
37, 48, G, 42, 43, 44	Upper Jurassic, lowest Cretaceous	6	335.2	45.6	323.8	46.5	65 330	8.4 3.7	5	255.7 58.4N
38ht, 41, 46, 40	Aptian–Albian	4	346.5	41.9	345.7	47.8	16 202	23.6 6.5	12	231.7 75.5N

No = sites number; G = study of [18]; N = number of measured samples; n = number of selected samples; Dg, Ig = characteristic direction before tectonic correction; Ds, Is = characteristic direction after tectonic correction; k = precision parameter; α_{95} = limit of the circle of 95% confidence. For the mean values, N = number of sites; ks/kg ratio of precision parameter after and before tectonic correction; Φ = longitude and λ = latitude of pole position associated to mean direction. Mean values for sites of the Jurassic and earliest Cretaceous are indistinguishable so we calculated the mean for the six sites.

LTC may be carried by pyrrhotite or it could result from a combination of several directions of magnetization. Consequently, we have not interpreted the LTC.

Site 41: The intensity of magnetization ranges between 10^{-5} and 10^{-4} A/m and 50–80% of the total magnetization is carried by goethite. The characteristic direction is defined between 300°C and 450°C and by alternative fields between 20 and 40 mT.

Site 46: The intensity of magnetization (10^{-3} A/m) decreases to 5×10^{-5} A/m after heating to 120°C. However, the magnetization remains well defined and the determination of the characteristic direction is unambiguous between 120°C and 520°C. For this lower Aptian site, 10 samples have normal polarities, whereas 1 is reversed and is antipodal to the others.

3.2.4. Albian (Figs. 5 and 6, Table 1)

Site 40: The NRM intensity is about $2 \times 10 - 5$ A/m. The characteristic direction of magnetization is determined between 200°C and 400°C.

4. Samples from the Lisbon area

To the North of Lisbon, 12 sites of Lower Cretaceous to Hauterivian were sampled in the region of Torres Vedras (Fig. 1). For most of the samples from this region, the NRM contains several components; however, we were unable to find coherence of the

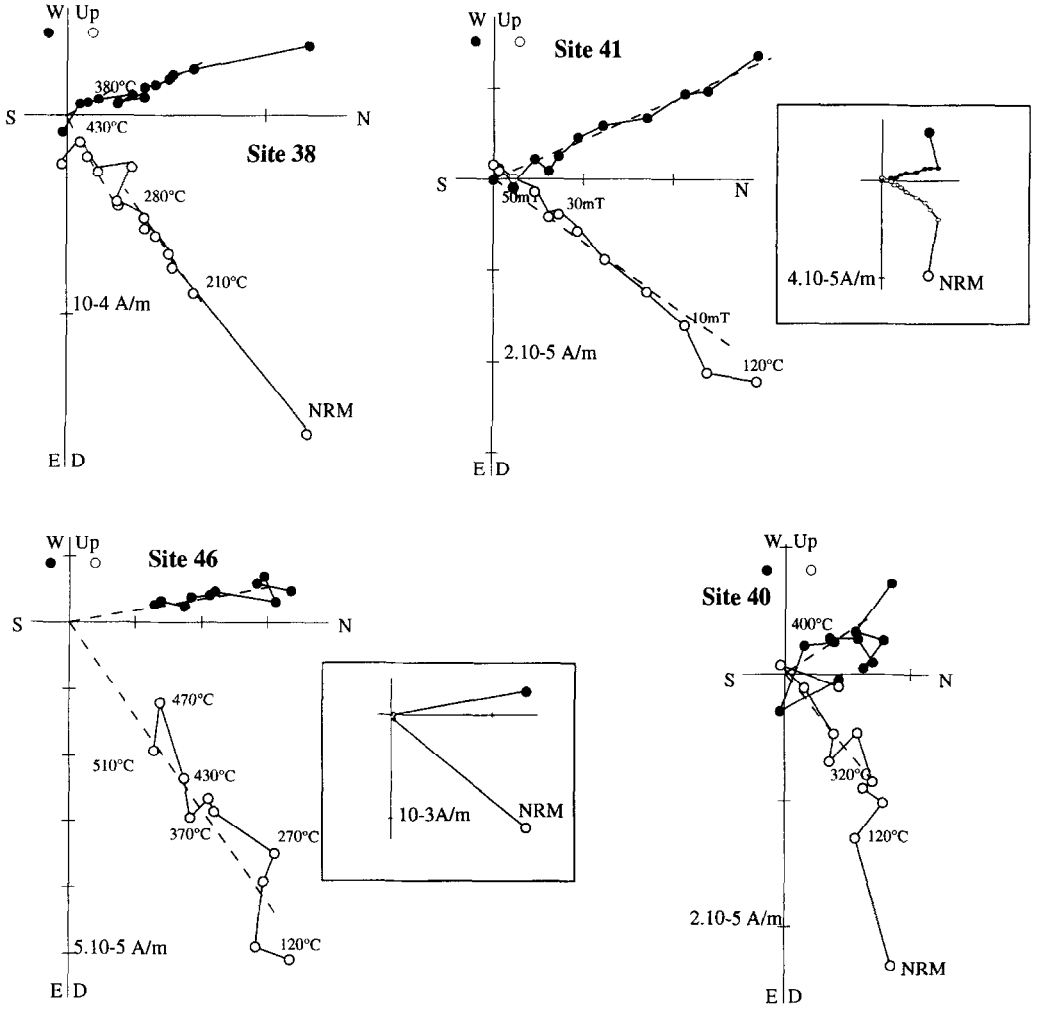


Fig. 5. Aptian–Albian sites, legend as in Fig. 3.

directions and could not concoct a reasonable interpretation of them.

In the Arrabida chain to the South of Lisbon, we sampled 7 sites of Jurassic and lowermost Cretaceous rocks. The magnetization intensities for most of these samples become too weak to measure above 120°C ($< 5 \times 10^{-6}$ A/m). Below that temperature, magnetization directions correspond either to the present field direction or are random.

In summary, we find the same problem encountered by [14] near the Sintra Massif west of Lisbon, in that the samples of these ages are remagnetized. Accordingly, we omit these samples from further consideration.

5. Paleomagnetic results

All results are listed in Table 1, together with the Berriasian data from Galbrun [18] from the oriental Algarve. Table 1 also lists the average directions that we calculated for the uppermost Jurassic (Portlandian) and for the earliest Cretaceous (Valanginian and Berriasian). Since the two Portlandian site mean

directions are indistinguishable from the earliest Cretaceous site mean directions, we combined all six site means to calculate a mean Portlandian to Valanginian direction (D = 324°, I = 46°, α_{95} = 4°). The fold test [22] for this set of sites is positive at the 95% confidence level. This result depends largely on the data from site 37 (Fig. 7a,b).

The precision parameter of the Aptian–Albian overall mean direction increases 12 times after a tectonic correction is applied (D = 346°, I = 48°, α_{95} = 6°) and gives a positive fold test at the 95% confidence level. Thus this direction predates Tertiary folding. Moreover, a reversed direction is observed in site 46 which is situated at the Barremian–Aptian boundary. Because a reversed interval of short duration is known to occur at that time, it is likely that this sample's magnetization was acquired during that interval, before the Cretaceous Long Normal magnetic superchron. We thus conclude that the pre-folding magnetization from these rocks may be considered a primary characteristic remanence.

6. Discussion

The calculated Portlandian–Valanginian paleomagnetic pole (Φ = 256°, λ = 58°, A_{95} = 5°) can be compared with previously published results. Before doing so, we note that a Barremian–Aptian remagnetization has occurred in the Iberian chain [13,15]. We consider that some data are remagnetized, such as that for the Sierra de los Cameros close to Soria [23]. Furthermore, we have not included the results from the Cantabric cordillera [24] because the age of the magnetization seems poorly constrained [23]. However, some rocks have survived complete remagnetization, such as the Oxfordian Aguilon and Tosos series situated at the edge of the Iberian basin [12,13]. Our paleomagnetic pole is in good agreement with the one obtained from those series. We also retained the paleomagnetic direction defined for the Hauterivian–Barremian by Galdeano [14] for sediments of the Lisbon region. Finally, we calculate for Iberia an average Malm–Hauterivian pole (ϕ = 261°, λ = 57°, A_{95} = 10°) (Table 2).

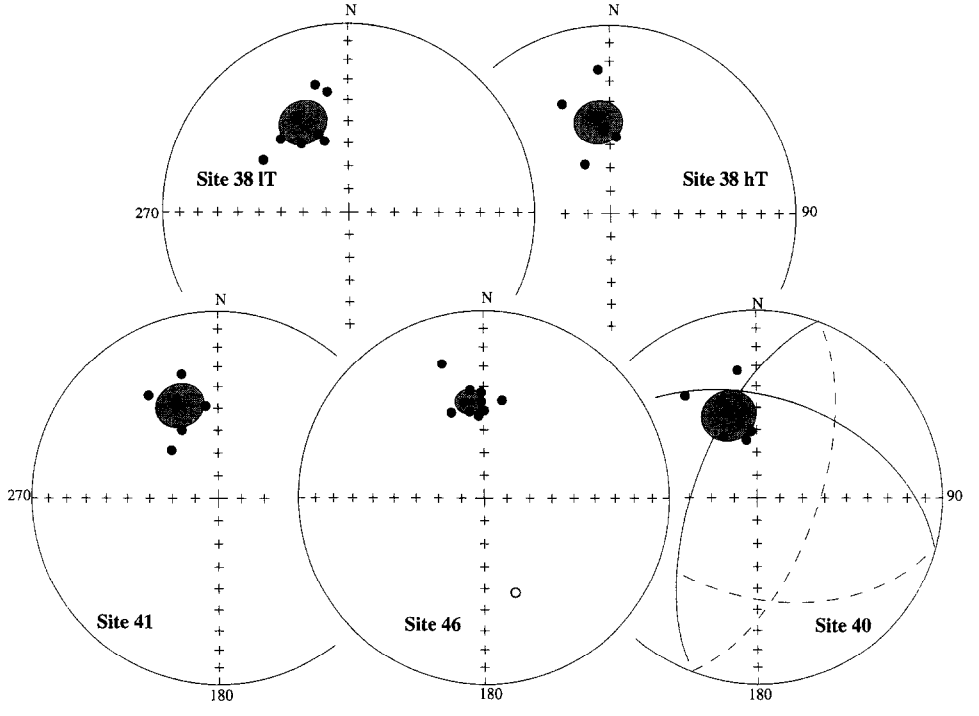


Fig. 6. Aptian–Albian sites; legend as in Fig. 4. For site 38 the two characteristic directions (lT = low temperature; hT = high temperature) have been drawn.

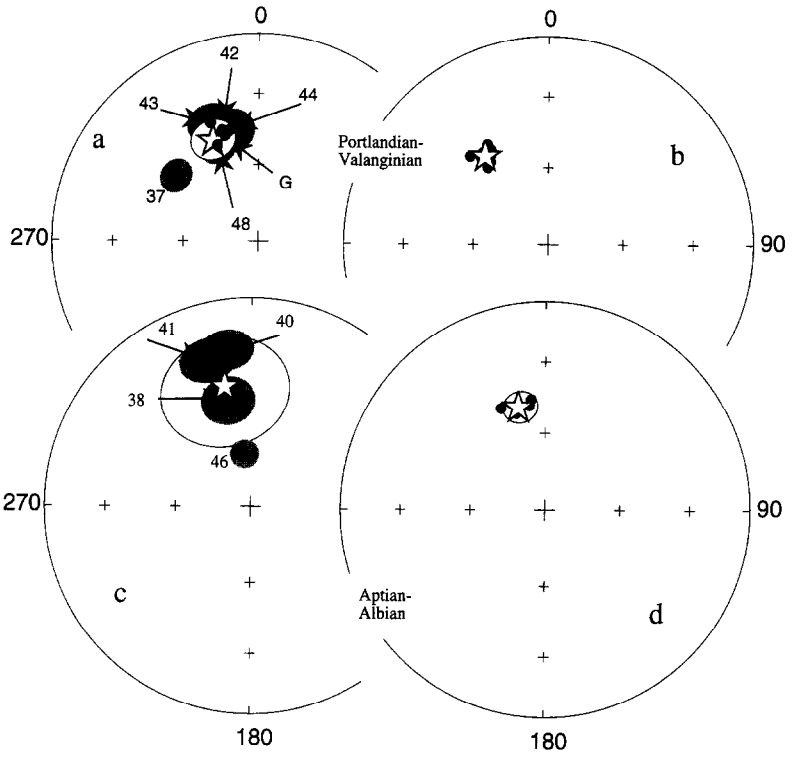


Fig. 7. Equal-area projection of characteristic directions (dots) and their mean (star with confidence ellipse). (a) Before tectonic correction and (b) after tectonic correction for Portlandian–Valanginian sites; (c) before tectonic correction and (d) after tectonic correction for Aptian and Albian sites.

Our paleomagnetic pole for the Aptian–Albian ($\Phi = 232°$ $\lambda = 75°$, $A_{95} = 8°$) is in good agreement with poles calculated from the Lisbon region [14], and from Maestrazgo [15]. From these data, we calculate an average pole for the Aptian–Albian of $\phi = 227°$, $\lambda = 74°$, $A_{95} = 5°$ (Table 2).

A comparison of the Portlandian–Hauterivian and Aptian–Albian pole position shows that Iberia has

Table 2
Paleomagnetic poles of Iberia (rounded to the degree)

Study	Age	N	Paleomagnetic poles		
			Φ	λ	A_{95}
Algarve, this study	Upper Jurassic/lowest Cretaceous		256°	58°	4°
Iberian Chain [13]	Oxfordian		251°	56°	6°
Lisbon area [14]	Hauterivian/Barremian		275°	57°	4°
Mean value	**Upper Jurassic/lowest Cretaceous 155–132 Ma**	**3**	**261°**	**57°**	**10°**
Algarve, this study	Aptian/Albian		232°	75°	6°
Iberian Chain [15]	Barremian, Aptian		225°	71°	7°
Lisbon area [14]	Aptian		226°	77°	7°
Mean value	**Barremian–Albian 124–97 Ma**	**3**	**227°**	**74°**	**5°**

Φ = longitude; λ = latitude; A_{95} = interval of confidence of the paleomagnetic poles, absolute age from [25].

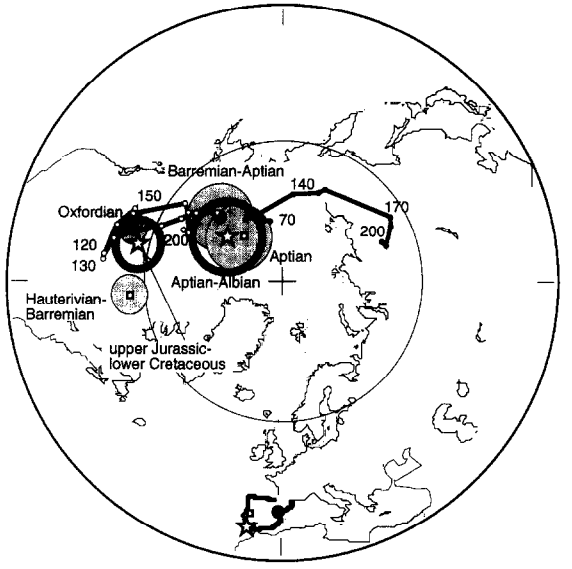


Fig. 8. Comparison of the curves of APWP for Eurasia (dots) and Africa (circles) between 200 and 70 Ma [27] with available poles for Iberia for the Upper Jurassic and lowest Cretaceous (Table 2): squares = Lisbon area, dots = Iberian chain; stars = the Algarve (poles and sites).

undergone a rotation of 22° ± 14°. Iberian poles for the Portlandian–Hauterivian are significantly remote from those of Eurasia (Fig. 8). In contrast, the position of our pole is situated very close to the coeval Eurasian pole for the Aptian–Albian. The main phase of relative rotation between Iberia and Eurasia would have begun during the Hauterivian–Barremian and ended sometime in the Barremian–Aptian. Thus, the rotation would have taken place mainly during the Barremian epoch which duration is approximately 8 Myr [25], whereas the rotation of Eurasia is negligible, 2° ± 9° [27]. The timing of the main rotation phase of Iberia [14,15] is better constrained than previously. This rapid motion of Iberia is well correlated with the major phase of extension in the Early Cretaceous. The extensional phase, which began around the end of the Valanginian to Hauterivian and was surely active during the Barremian–Aptian, is documented by the development of normal faults on the continental margins both west and north of Iberia [26]. The slight chronological shift between paleomagnetic and geological data is not significant because the extension may have begun before the

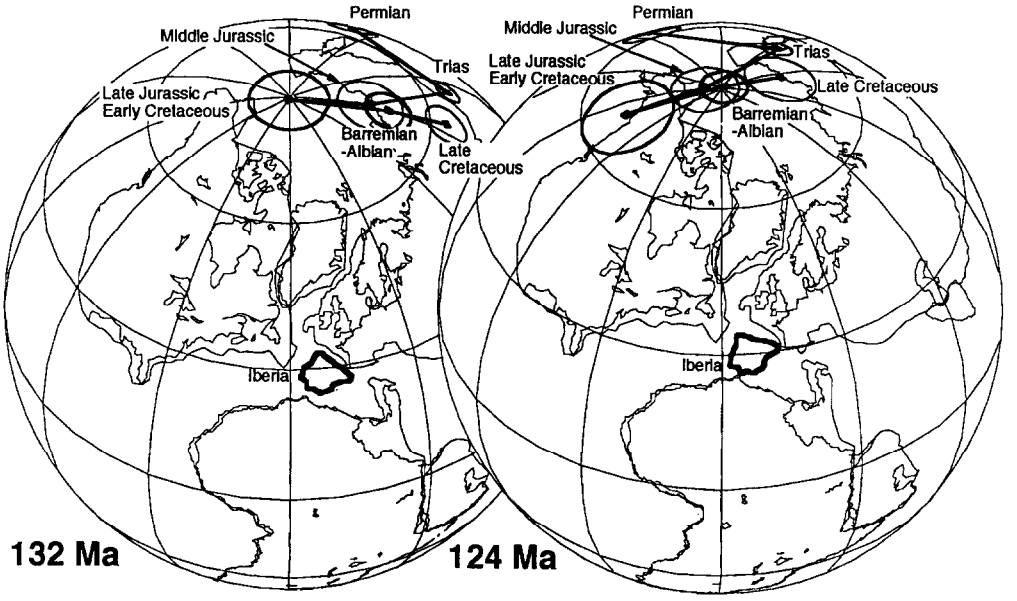


Fig. 9. Iberia and its APWP plotted in a global frame based on paleomagnetic data and kinematic reconstruction of major plates [1,27]. The following paleomagnetic poles for Iberia have been used: Late Cretaceous [29,30]; Barremian–Albian, this study; Late Jurassic–Early Cretaceous, this study; Middle Jurassic [31]; Trias [7]; Permian [7,29]. By 124 Ma Iberia had achieved a rapid rotation. By 132 Ma Iberia was probably attached to Eurasia.

movement of rotation was significant enough to be recorded by the paleomagnetic data.

7. Conclusion

A plate tectonic reconstruction of Africa and Europe based on the paleomagnetic data base of [27] and oceanic magnetic anomalies and tectonic lineations [1] is shown in Fig. 9. The APWP attached to the Iberian plate is also shown in Fig. 9. In order to evaluate the Iberian location at a given time, first the corresponding paleomagnetic pole is put on the Earth' axis, and then a rotation about a constant line of latitude is made by eye. This approach is limited by the use of the present Iberian coastline (instead of reconstructed plate boundaries) as well as uncertainties associated with the reconstruction of the major plates.

At 132 Ma (Fig. 9), Iberia is well located in relation to the major plates with its present west coast aligned parallel to the Grand Banks and its present northern margin parallel to the border of the Eurasian shelf. Iberia was still attached to Eurasia, while Africa was moving southeastwards with respect to North America and Eurasia.

We also show a reconstruction of these plates at 124 Ma (Fig. 9). It is clear from the comparison of the two reconstructions that Iberia accomplished a significant rotation between the two times (132 and 124 Ma), as suggested by our results. Fig. 9 (124 Ma) suggests that Iberia may have been dragged by Africa. At this stage, Iberia follows, but without being a real part of, the African plate. In this reconstruction, the space between Iberia and Africa appears narrower than in the 132 Ma reconstruction. Although this apparent difference may result from the Iberian boundaries we used or from uncertainties in the paleomagnetic and/or kinematic reconstructions, a confining of Iberia between Africa and Eurasia would provide suitable conditions for the observed rapid rotation of this microplate.

The quick rotation of Iberia during the Early Cretaceous was preceded and followed by small amplitude rotations with respect to both Eurasia and Africa [11]. This does not exclude the possibility of significant relative motions, since paleomagnetic measurements poorly constrain "translatory" movements (rotation around a distant Eulerian pole inducing a very low variation of the declination), especially if the variation in latitude is limited.

Thus, before the Hauterivian, Iberia was attached to Eurasia, while Africa separated from America. During the Barremian, Iberia underwent a quick rotation with respect to Eurasia. This rotation event is contemporaneous with other tectonic events occurring in the western Tethyan realm [28] and is contemporaneous with the beginning of the "hair-pin" turn in the APWP curves of Europe and Africa [27], possibly suggesting a single cause for the change in kinematics. After this episode of rotation, it seems that Iberia followed Africa in its eastward motion and experienced a moderate rotation with respect to Eurasia, while oceanic spreading was under way in the Bay of Biscay.

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